

Foundation of Blockchain Technology (IS501) **Pre-Reading Material – Lecture 5**

Proof of Stake

Estimated Reading Time: 20–25 minutes

Estimated Reading Time: 15–20 minutes

Course Context

In previous lectures, we studied **Proof-of-Work (PoW)** and its role in achieving consensus in blockchain systems. This lecture introduces **Proof-of-Stake (PoS)** as an alternative approach.

The key question is:

Can blockchain consensus be achieved without requiring large amounts of computational work?

Proof-of-Stake replaces the computational resource used in PoW with an **economic stake** in the blockchain.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the purpose of Proof-of-Work.
- Identify major limitations of PoW.
- Explain the basic concept of Proof-of-Stake.
- Compare PoW and PoS.
- Identify important security challenges in PoS.
- Explain the advantages of PoS.



Activate Your Prior Knowledge

Recall from the previous lecture:

- Bitcoin uses **Proof-of-Work** for consensus.
- Miners perform computational work.
- Mining requires computational resources and energy.
- PoW makes rewriting blockchain history computationally expensive.

Think About It

If computational work is used to secure a blockchain:

Could economic ownership of the blockchain's cryptocurrency be used instead of computational power?

Core Concepts

1. Why Proof-of-Work?

Proof-of-Work serves two important purposes.

Sybil Attack Resistance

PoW requires participants to use **computing resources** to create blocks. Therefore, simply creating a large number of identities does not provide an attacker with unlimited influence.

"Arrow of Time"

The computational work required to create blocks limits how quickly blocks can be generated. This helps maintain the ordering of blocks and makes certain attacks, such as double-spending, more difficult.



2. Limitations of Proof-of-Work

High Energy Consumption

Mining requires a large number of computational operations. As the competition among miners increases, substantial computing resources and energy may be consumed.

Centralization

Although anyone can participate in mining, large mining operations can benefit from economies of scale. Access to specialized hardware, inexpensive electricity, and large infrastructure can lead to concentration of mining power.

These limitations motivate the study of alternative consensus mechanisms.

3. What Is Proof-of-Stake?

Proof-of-Stake (PoS) is a consensus mechanism in which a participant's influence is related to the amount of cryptocurrency they have **staked**.

A simple way to compare the two mechanisms is:

Proof-of-Work

Computational power influences participation

Miners

Requires computational work

High energy requirement

Security based on cost of computation

Proof-of-Stake

Economic stake influences participation

Stakers/validators

Does not depend on PoW-style mining

Lower computational requirement

Security based on economic stake

The basic idea is that an attacker would need to acquire a significant amount of the cryptocurrency in order to obtain substantial influence over the consensus process.



4. Security Challenges in Proof-of-Stake

Replacing PoW with PoS does not eliminate security problems. Instead, it creates different challenges.

Grinding Attack

A participant may attempt to manipulate possible choices or randomness in order to increase the probability of being selected to produce a block. This is known as a **grinding attack**.

Prediction Attack

If the next block producer can be predicted, an attacker may have an opportunity to target, bribe, or otherwise influence that participant.

History Revision Attack

PoW makes rewriting blockchain history expensive because the attacker must reproduce the computational work.

PoS does not rely on the same type of accumulated computational work, so protocols need other mechanisms to discourage attempts to rewrite history.

5. Advantages of Proof-of-Stake

Faster Confirmation

Some PoS systems can confirm transactions significantly faster than traditional PoW systems such as Bitcoin.

Lower Computational Requirement

PoS does not require miners to continuously perform large amounts of hash computation to compete for blocks.

This can significantly reduce the computational and energy requirements associated with consensus.



Accountability

PoS participants have an economic stake in the system.

If a protocol identifies a participant as violating its rules, the participant's stake can potentially be **penalized or taken away**. This provides an additional mechanism for discouraging dishonest behavior.

PoW vs. PoS: At a Glance

Feature	Proof-of-Work	Proof-of-Stake
Participant	Miner	Staker/Validator
Main resource	Computing power	Cryptocurrency stake
Energy requirement	High	Lower
Block selection	Based on PoW competition	Based on stake/protocol rules
Main security cost	Computational cost	Economic cost
Major concern	Energy & mining centralization	Different attack vectors
Accountability	Limited direct financial penalty	Stake can potentially be penalized

Guided Reading Questions

1. Why does Bitcoin use Proof-of-Work?
2. What are the two major limitations of PoW discussed in this lecture?
3. What is the basic idea behind Proof-of-Stake?
4. How does PoS differ from PoW in terms of the resource used for consensus?
5. What is a grinding attack?
6. Why can predictability be a security concern in PoS?
7. What is one important advantage of PoS over PoW?



Reflection Activity

Choosing a Consensus Mechanism

Imagine you are designing a blockchain for a large financial application.

You have two choices:

Proof-of-Work:

Strong computational security but high computational and energy requirements.

Proof-of-Stake:

Lower computational requirements and potentially faster confirmation, but introduces different security challenges.

Question:

Which mechanism would you choose and why?

Write **100–150 words** explaining your choice and the trade-off you would accept.

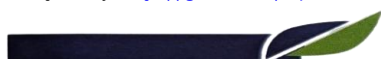
Self-Assessment

Fill in the blanks

1. Bitcoin uses _____ as its consensus mechanism.
2. Proof-of-Stake uses economic _____ as an important basis for participation.
3. Participants in a PoS system are commonly called _____ or validators.
4. An attempt to manipulate possible choices to improve the probability of block selection is called a _____ attack.
5. PoS can provide faster transaction _____ than traditional PoW systems.

True or False

6. Proof-of-Stake completely eliminates blockchain security problems.
7. Proof-of-Work can require significant computational resources.
8. Proof-of-Stake can reduce the computational requirements of consensus.



9. PoS introduces security challenges that are different from those of PoW.

Key Takeaways

- **Proof-of-Work** uses computational work as the basis for blockchain consensus.
- Major PoW concerns include **energy consumption and mining centralization**.
- **Proof-of-Stake** uses economic stake rather than computational competition.
- PoS can provide **lower computational requirements, faster confirmation, and economic accountability**.
- PoS introduces its own challenges, including **grinding, prediction, and history-revision attacks**.
- Neither PoW nor PoS is simply "better"; each involves different **security, economic, and performance trade-offs**.

Remember

Proof-of-Work → Security through computational cost

Proof-of-Stake → Security through economic stake

